

# **Generator Differential Protection (ANSI 87G) Siprtotec 5 7UM85**

Md Kamruzzaman Prodhan

Engineer, Rooppur Nuclear Power Plan Project

Reference: Manual of SIPROTEC 5 Generator Protection 7UM85 V9.5

# Generator Differential Protection(ANSI 87G)

## Overview of Functions:

- Detects short circuits in generators.
- Detects ground faults in generators with a grounded neutral point (typical for low voltage).
- Stable operation in the event of external short circuits with current transformer saturation through intelligent saturation detection methods.
- Triggers safely in the case of internal high-current faults through an additional high-current stage.

# Generator Differential Protection(ANSI 87G)

## Structure of the Function:

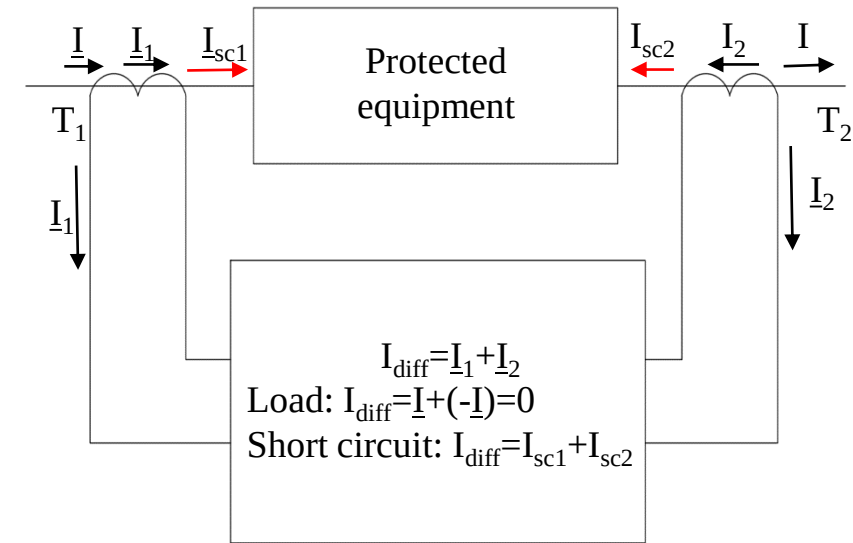
The **Generator differential protection** function is used in the **Generator Diff** protection function group. It function consists of the 3 tripping stages:

1. I-DIFF
2. I-DIFF fast
3. I-DIFF unrestrained

# Generator Differential Protection(ANSI 87G)

## Basic Principle of Differential Protection:

- Differential protection works on the principle of Kirchhoff's Current Law "The algebraic sum of currents at a node is zero."
- For a generator the current entering is equal to the current leaving it during normal operation or during an external fault.
- CTs are installed at both ends of the protected zone. Their secondary currents are compared.
- If the vector sum of currents is zero, the relay remains stable.
- If there is a difference, it indicates an internal fault.

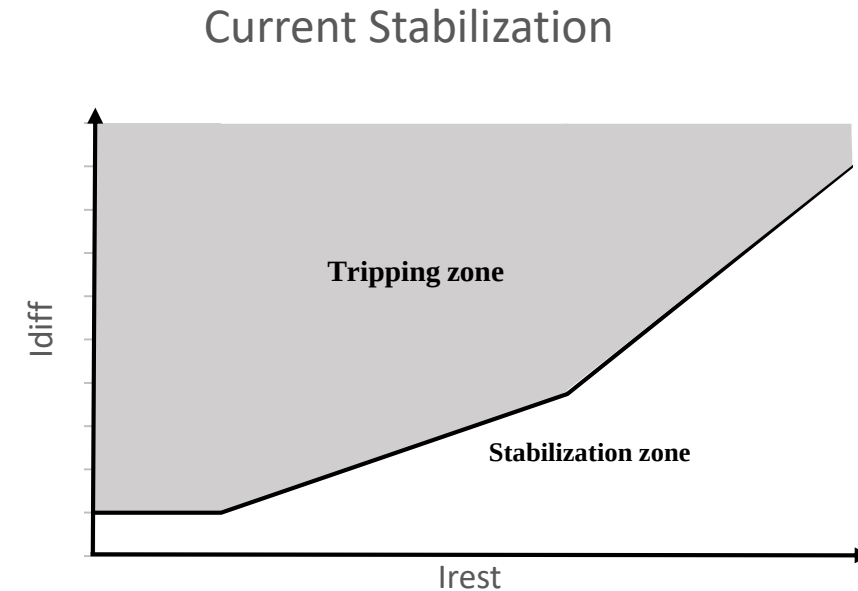


# Generator Differential Protection(ANSI 87G)

## Current Stabilization:

If very large through-current flows due to external fault, differences in CT characteristics/CT saturation may create a false differential current,  $I_{diff}$ . If  $I_{diff}$  exceeds the pickup threshold, the relay might trip incorrectly, even though the fault is outside the protected zone.

To prevent this, restraining current  $I_{rest}$  is introduced.  $I_{rest}$  is maximum value of the two CT currents ( $I_{rest} = \text{Max}(|I_1|; |I_2|)$ ). The relationship is defined by a **characteristic curve**:  $I_{diff} = f(I_{rest})$ . This is a stabilizing reference to distinguish external faults from internal faults.

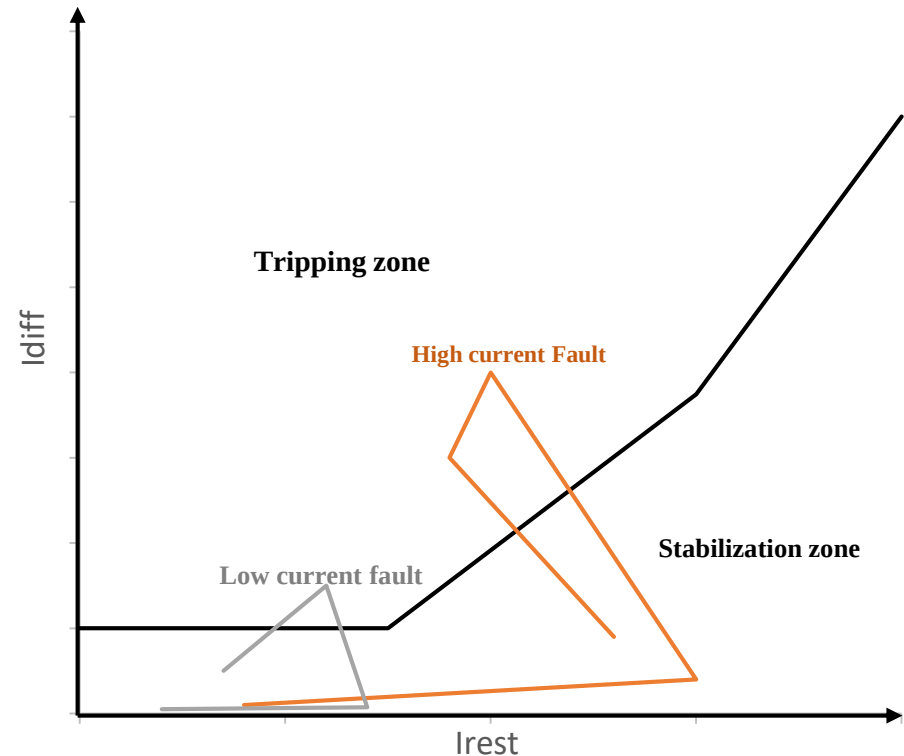


# Generator Differential Protection(ANSI 87G)

## Additional Stabilization:

**High-current external short circuit:** At the initial stage both CTs act with the fault current correctly and  $I_{rest}$  is high,  $I_{diff}$  is near zero. After this when the CTs become saturated due to high current, it delivers a distorted or clipped secondary current. This creates a **false**  $I_{diff}$ , which may cross the relay's operating curve. Simultaneously, the  $I_{rest}$  drops. Once the transformer is no longer saturated, the differential current decreases and falls below the characteristic curve.

Additional Current Stabilization



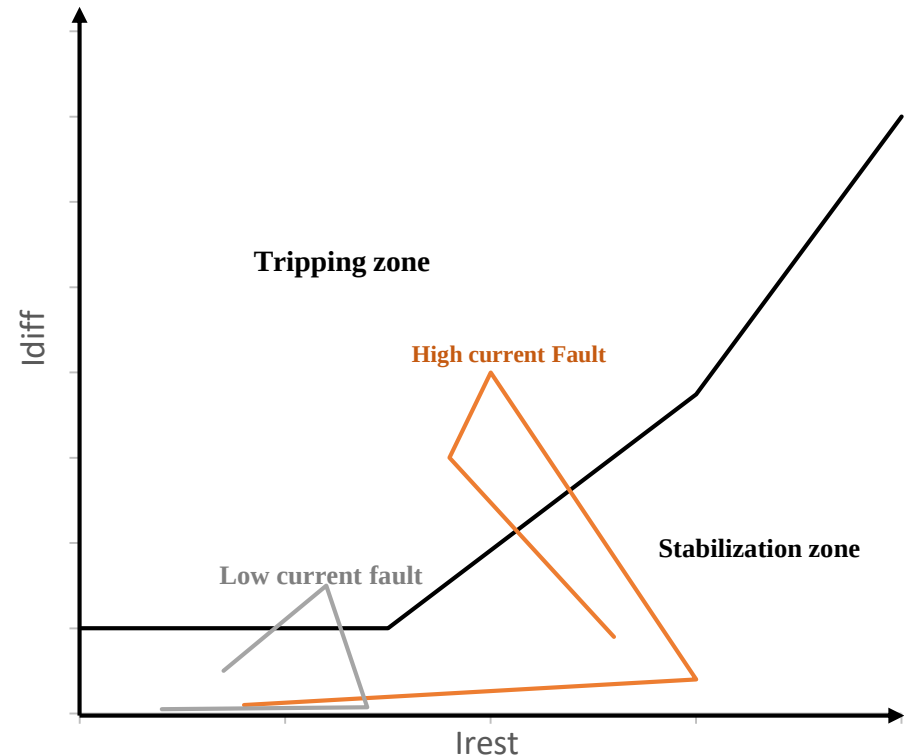
**Intelligent saturation-detection methods capture this state and lead to a temporally limited blocking of the Differential protection function.**

# Generator Differential Protection(ANSI 87G)

## Additional Stabilization:

**Low-current, external short circuit:** CTs can also become saturated through the large direct-current time constant (switching of transformers/motors) but small currents flowing through. This manifests itself as a phase angle rotation of the current. Exceeding the characteristic curve in a non-stabilized area is possible.

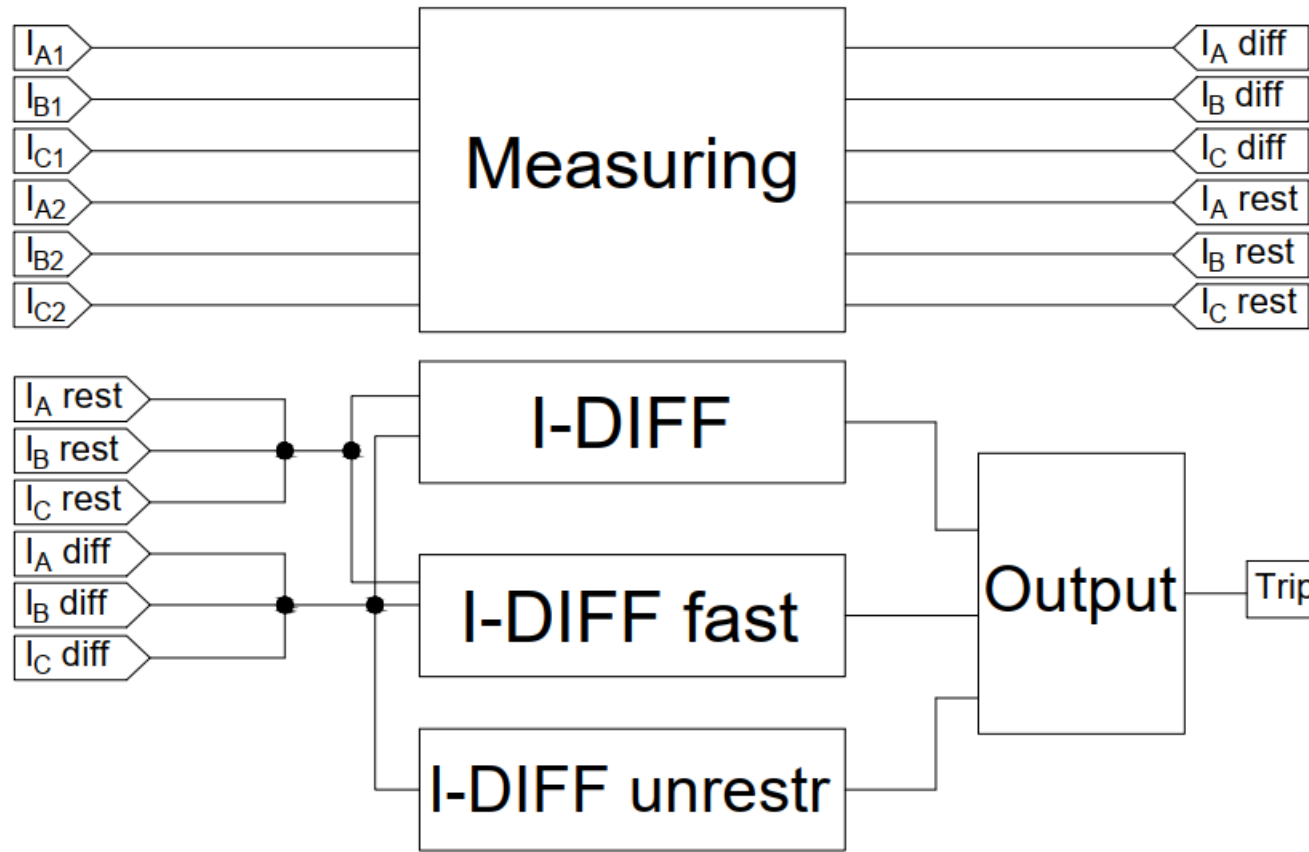
Additional Current Stabilization



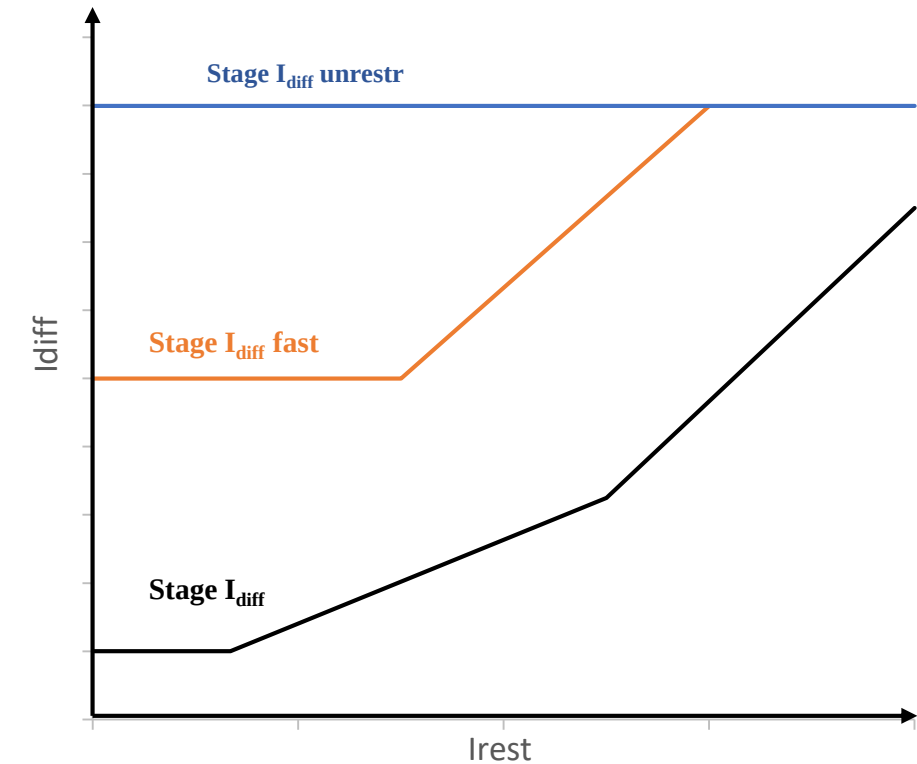
**Intelligent saturation-detection methods capture this state and lead to a temporally limited blocking of the Differential protection function.**

# Generator Differential Protection(ANSI 87G)

## Logic of the Function



Logic Diagram



Operative Curve

# **Generator Differential Protection(ANSI 87G)**

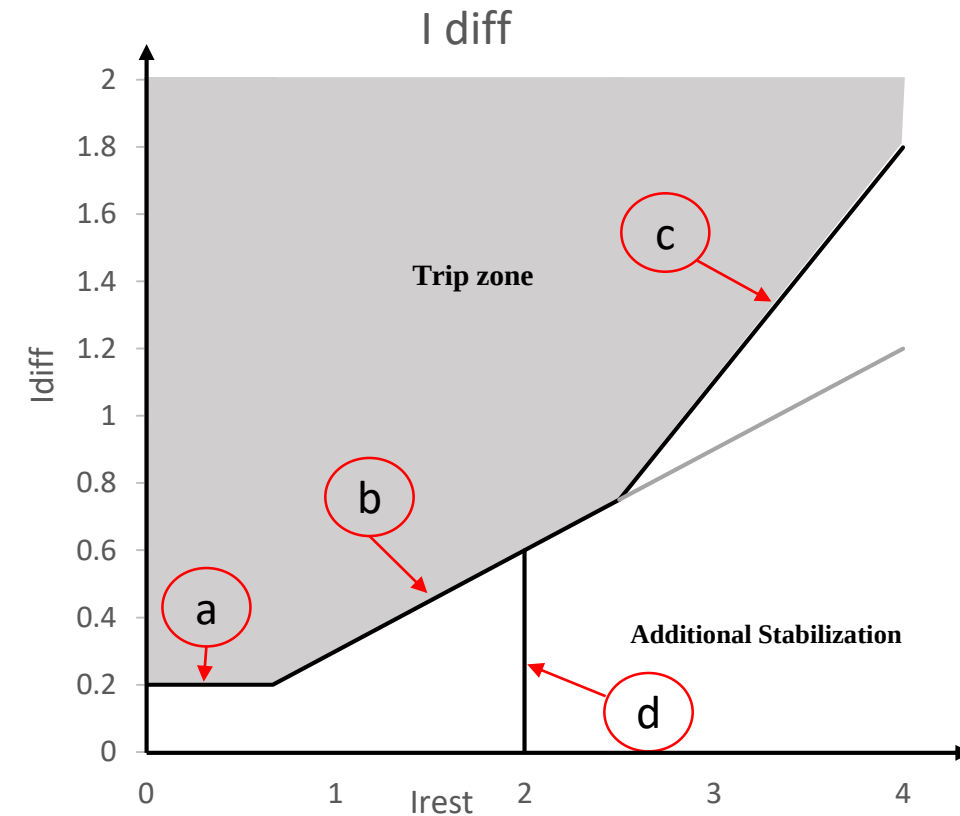
## **I-DIFF**

- 1. Pickup Characteristic.**
- 2. Detection of External Faults.**
  - 1. Blocking in the case of a definite high-current external fault.**
  - 2. Removal of the blocking with a change in fault location from external to internal.**
- 3. DC Component Detection.**
- 4. Startup Detection.**

# Generator Differential Protection(ANSI 87G)

## I-DIFF(1. Pickup Characteristic):

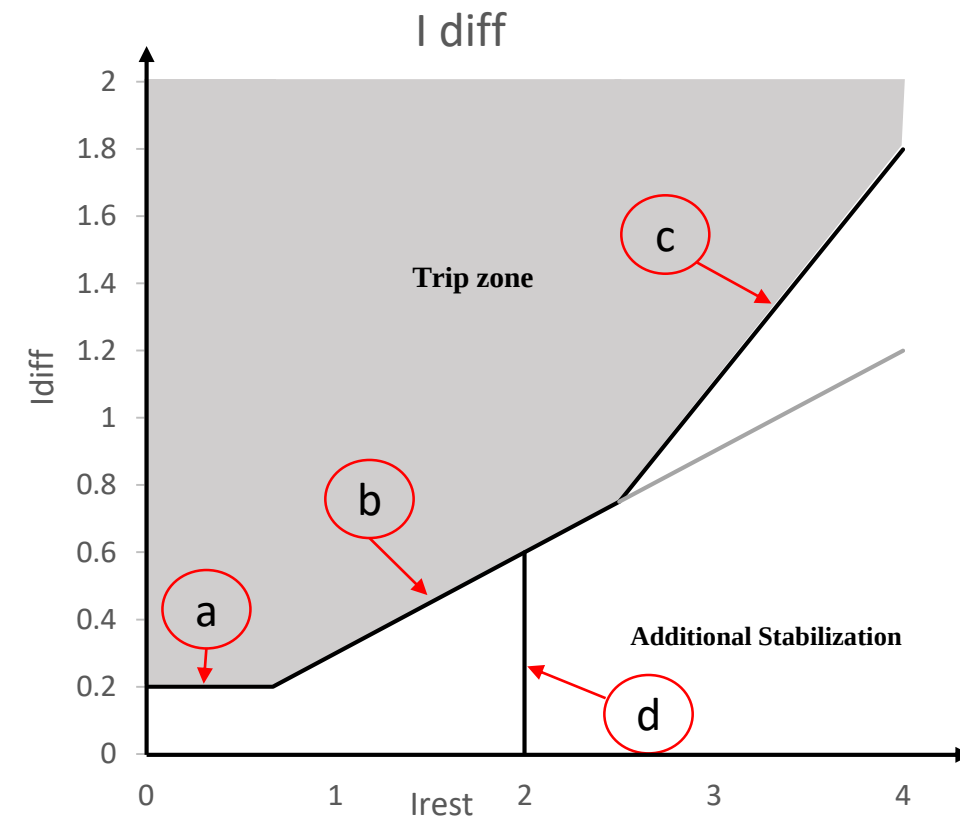
- The characteristic curve branch **a** represents the sensitivity threshold of the differential protection (parameter **Threshold**) and considers constant fault currents like magnetizing currents.
- The characteristic branch **b** considers current proportional faults under normal operating conditions. These faults can occur due to errors in the current transformer, the input transformer of the device and Adaptation deviations.
- The characteristic curve branch **c** protects the differential protection against over function in case of high current due to external faults and transformer saturation.



# Generator Differential Protection(ANSI 87G)

## I-DIFF(2. Detection of External Faults ):

- When the  $I_{rest}$  current is greater than  $d$  (Threshold add on stabiliz) and  $I_{diff}$  is less than slope of  $b$ , it detect this as external fault and block differential protection. When fault clear and  $I_{rest}$  falls below 0.7 times of Threshold add-on stabiliz( $d$ ), blocking reset. If the  $I_{diff}$  exceeds Slope  $b$  as a result of CT saturation, the blocking is limited in time by parameter: Time of add-on stabiliz in setting value.
- If in this condition an internal fault occurs. The relay analyzes instantaneous values of  $I_{diff}$ , during saturation,  $I_{diff}$  values dip below an internal threshold. If  $I_{diff}$  permanently exceeds this threshold while the phase is already in pickup, the relay interprets it as an internal fault and remove blocking.



# Generator Differential Protection(ANSI 87G)

## I-DIFF(3. DC Component Detection):

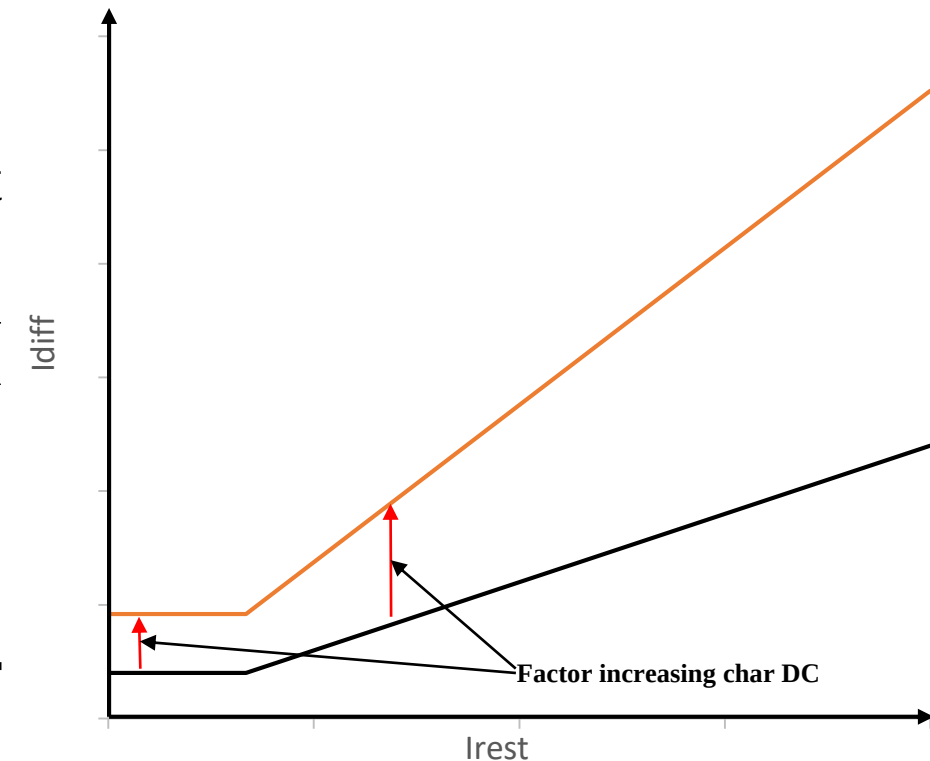
The presence of DC component is determined by 3 conditions.

1. The relay monitors  $I_{rest}$  for sudden jumps.
2. After a 1-period delay, the DC component  $I_{rest.DC,phsx}$  is calculated from the instantaneous current of the phase with the highest current and compared with  $I_{rest,phsx}$  against an internal threshold.
3.  $I_{diff}$  in all 3 phases must be below  $0.7 \times$  characteristic curve.

If conditions are satisfied, the operate characteristic is raised. Multiply slope  $b$  and threshold by factor increasing char but not more than 0.85.

The relay measures how long the DC offset persisted, then waits the same amount of time plus 5 cycles before restoring normal sensitivity.

Raising Operating curve for DC detection



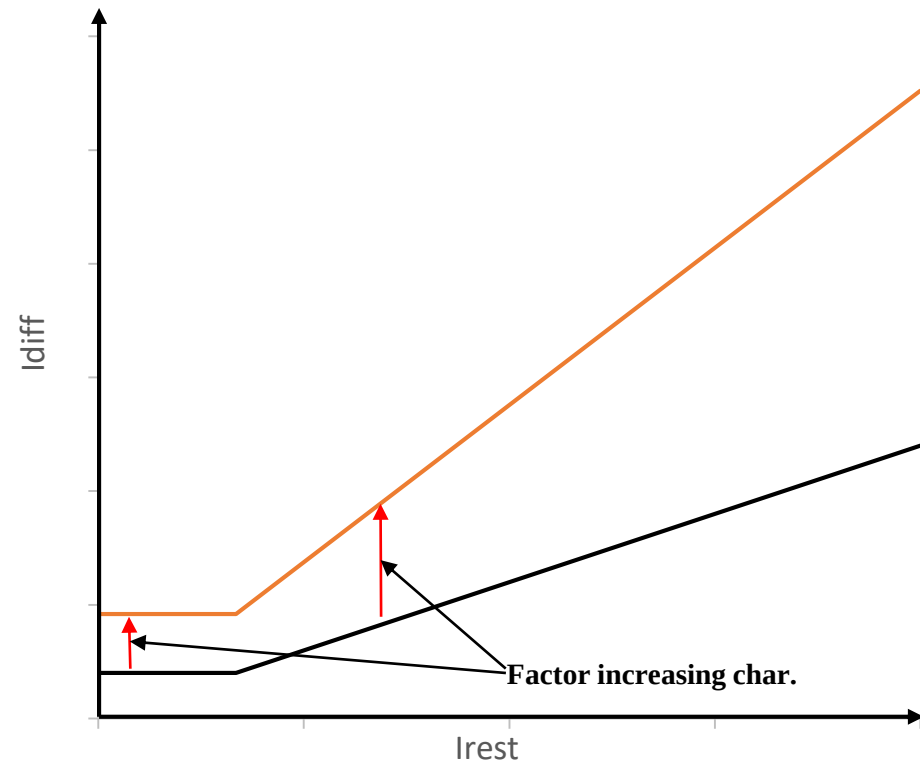
# Generator Differential Protection(ANSI 87G)

## I-DIFF(4. Startup Detection):

SIPROTEC monitors  $I_{rest}$  in all 3 phases. If  $I_{rest}$  is less than **Thresh.** **startup detection**, the relay assumes generator is in startup mode. After detection of startup The pickup value Threshold and Slope b are raised by **Factor increasing char.** specified in setting. Slope b cannot exceed a value of 0.85.

The operating curve restore to normal after expiration of time **delay(Max. perm. Start time)** specified in the setting.

Raising Operating curve for Startup Detection



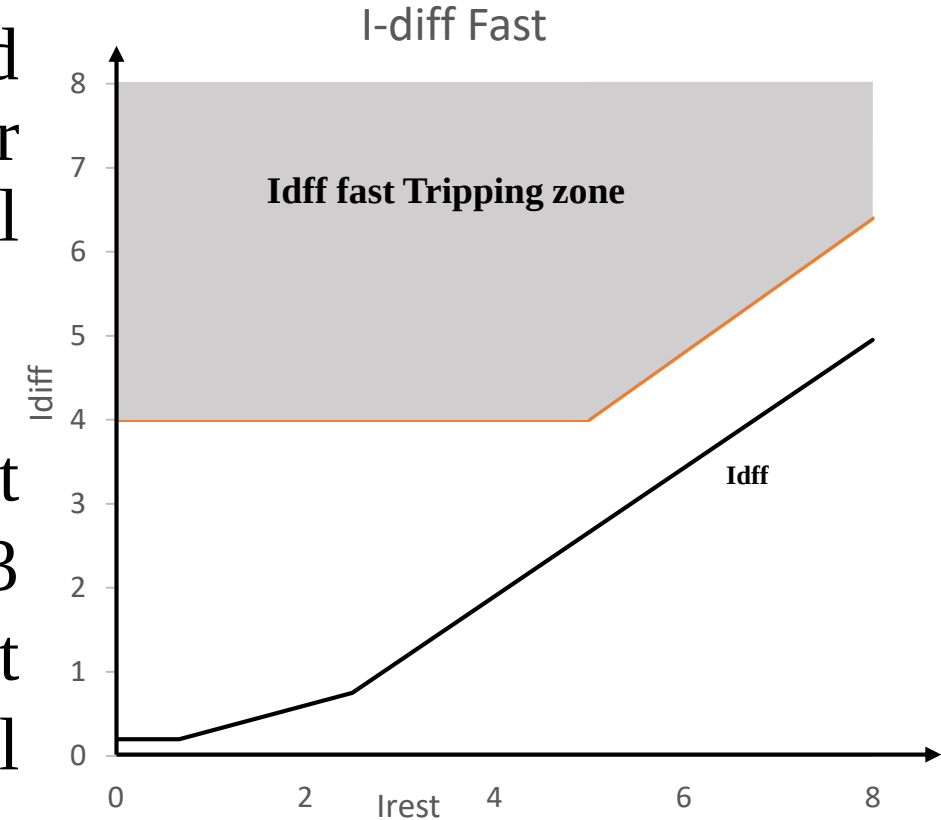
# Generator Differential Protection(ANSI 87G)

## I-DIFF Fast:

This is a supplementary stage to the standard restrained I-DIFF. Its main job is to trip faster for high current internal faults than the normal I-DIFF and it stays stable in external faults.

### Instantaneous Value Method:

Works on sampled values of differential current  $idiff(t)$  and restraint current  $irest(t)$ . There are 3 conditions. (1)Monitoring restraint current derivative ( $direst/dt$ ) and compare with internal threshold.



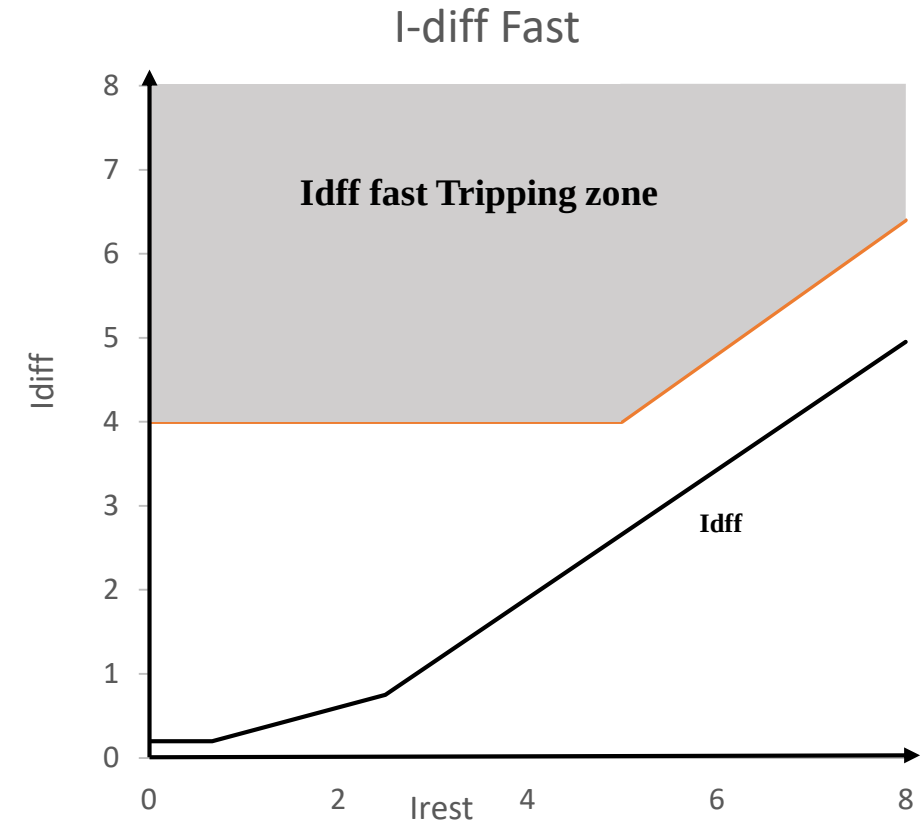
# Generator Differential Protection(ANSI 87G)

(2) Differential current exceeds slope line  $idiff(t) > 0.8 * irest(t)$ . (3) Differential current is greater than threshold.

If all three conditions are true trip immediately. If condition 1 but not condition 2 are true, block function for 300 ms (stability for external faults with CT saturation).

## Fundamental Component Method:

Processes the same measurements as normal I-DIFF, but with slight filtering delay. Compares  $Idiff$  (fundamental) vs.  $Irest$  (fundamental) with slope = 0.8.



# Generator Differential Protection(ANSI 87G)

## I-DIFF Unrestrained:

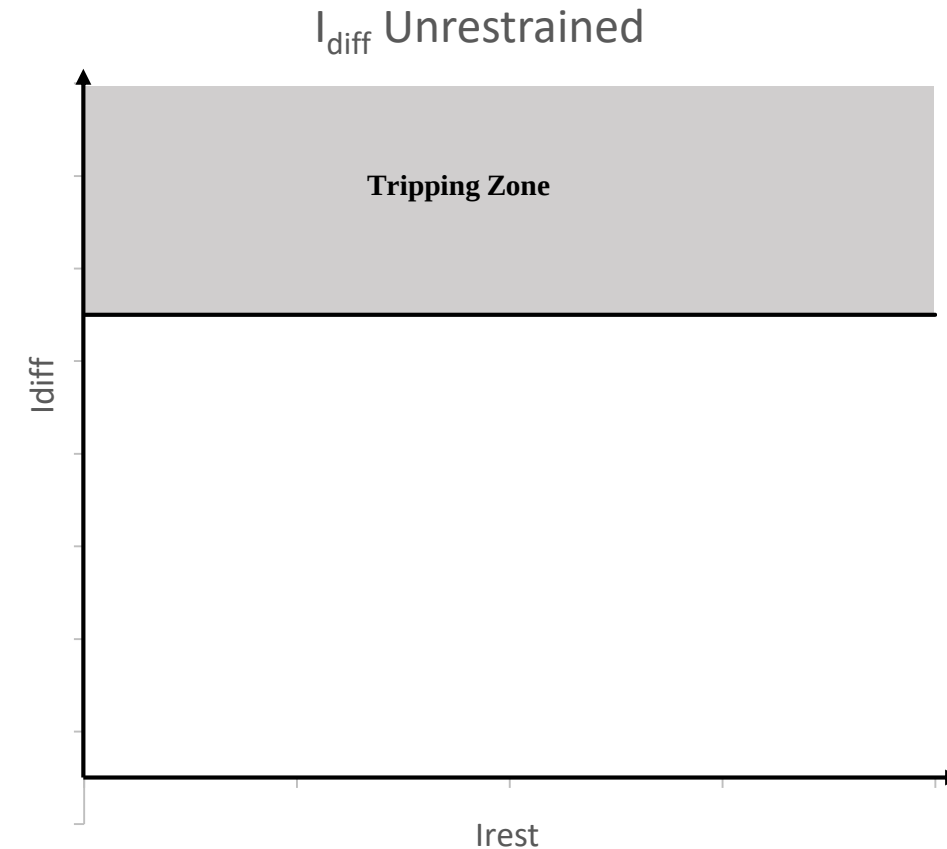
It trips instantly on very high internal fault currents and does not depend on  $I_{rest}$  ensuring CT saturation or harmonics don't cause false blocking. There are two methods of detection.

### 1. Instantaneous-value method:

works directly on the sampled differential current  $I_{diff}(t)$  and compare with set value for trip.

### 2. Fundamental-component method:

Extracts the fundamental component of the differential current with slight time delay due to filtering. Trip decision is based on the fundamental operate curve.



# Generator Differential Protection(ANSI 87G)

| Comparison of Tripping Stages |                                                                         |                                                                |                                                  |                    |
|-------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------|--------------------|
| Function                      | Logic                                                                   | Main Use                                                       | CT Saturation Handling                           | Speed              |
| I-DIFF                        | Idiff vs Irest with slope characteristics                               | General differential protection                                | Stable, inrush blocking                          | Normal             |
| I-DIFF Fast                   | Instantaneous & Fundamental, Check $I_{diff} > 0.8 I_{rest}$ +threshold | High current internal faults with stability in external faults | Unrestrain derivative of Irest for CT saturation | Faster than I=DIFF |
| I-DIFF Unrestrained           | Idiff only, no restraint & instant trip                                 | Extremely high internal faults(above external limit)           | No need                                          | Fastest            |